SYSTEM OVERVIEW

The Micromodule-85, hereafter referred to as MM-85, is an 8085 based microprocessor trainer. It has a 24-key keyboard, a 6-digit display, and 3 programmable input/output registers. Only a 5 volt power source capable of delivering .5 Amp is required to operate the MM-85. The MM-85 kit can be assembled in as little as 2 hours.

The MM-85 has the capability of performing the following operations:

- Writing program steps or data into RAM
- Reading the contents of memory
- Executing programs
- Single stepping through the execution of a program and examine the registers at any point
- Setting up to three breakpoints
- Setting user definable software interrupt vectors

MM-85 SPECS

CENTRAL PROCESSOR

CPU: 8085A
Instruction Cycle: 1.3 μsecond
Tcy: 330ns

MEMORY

EPROM: 8755 2K-bytes 0000-07FF Monitor Program
RAM: 8155 256-bytes 0800-08FF Monitor area
8185 1K-bytes 1400-17FF User area

INPUT/OUTPUT (I/O Connector)

Parallel: 8155 22 lines
Serial: SID/SOD on the 8085
Interrupts: RST 7.5
RST 6.5
INTR
Power: Vcc & Vss

SOFTWARE

Monitor: Preprogrammed 8755

ASSEMBLY INSTRUCTIONS

Organize

Before starting assembly, it's a good idea to organize your workplace. You should have room to accommodate this document, your tools and soldering iron, the circuit board, and the various parts.

The following tools will be needed to assemble the MM-85:

Needle-nose pliers
Small diagonal cutters
Screwdriver
Soldering iron (<35 watts) with a small tip
Rosin-core solder

As you unpack the parts you should identify each part and check it off the parts list. Don't handle the devices shipped with protective anti-static material; static discharge may damage these MOS devices.

Report any missing parts before assembly begins.

Assembly Techniques

You should follow these steps to produce a professional quality circuit board:

1. Measure each component's fit. Bend all component leads with your needle nose pliers carefully to make the part fit exactly the way you want it to and no undue stress is placed on the component.

2. Orient each component so that any value designations may be read easily. Bend the leads slightly to hold the component in place.

3. Do not trim the leads before soldering.

4. Do not solder components until you have been instructed to do so. Solder on the bottom of the PC board only.

5. Be sure your soldering iron is clean and tinned. The iron point should contact the PC board and the lead simultaneously. Touch the opposite side of the lead with the solder only long enough to melt a very small amount of solder and allow it to flow around the lead and onto the pad. Be careful not to let solder flow to adjacent pads.

6. If any solder connection is not shiny with solder spread smoothly around it, then reheat with the iron. It should not need any more solder.

7. Clip of excess leads as close as possible to the board after it has been soldered.

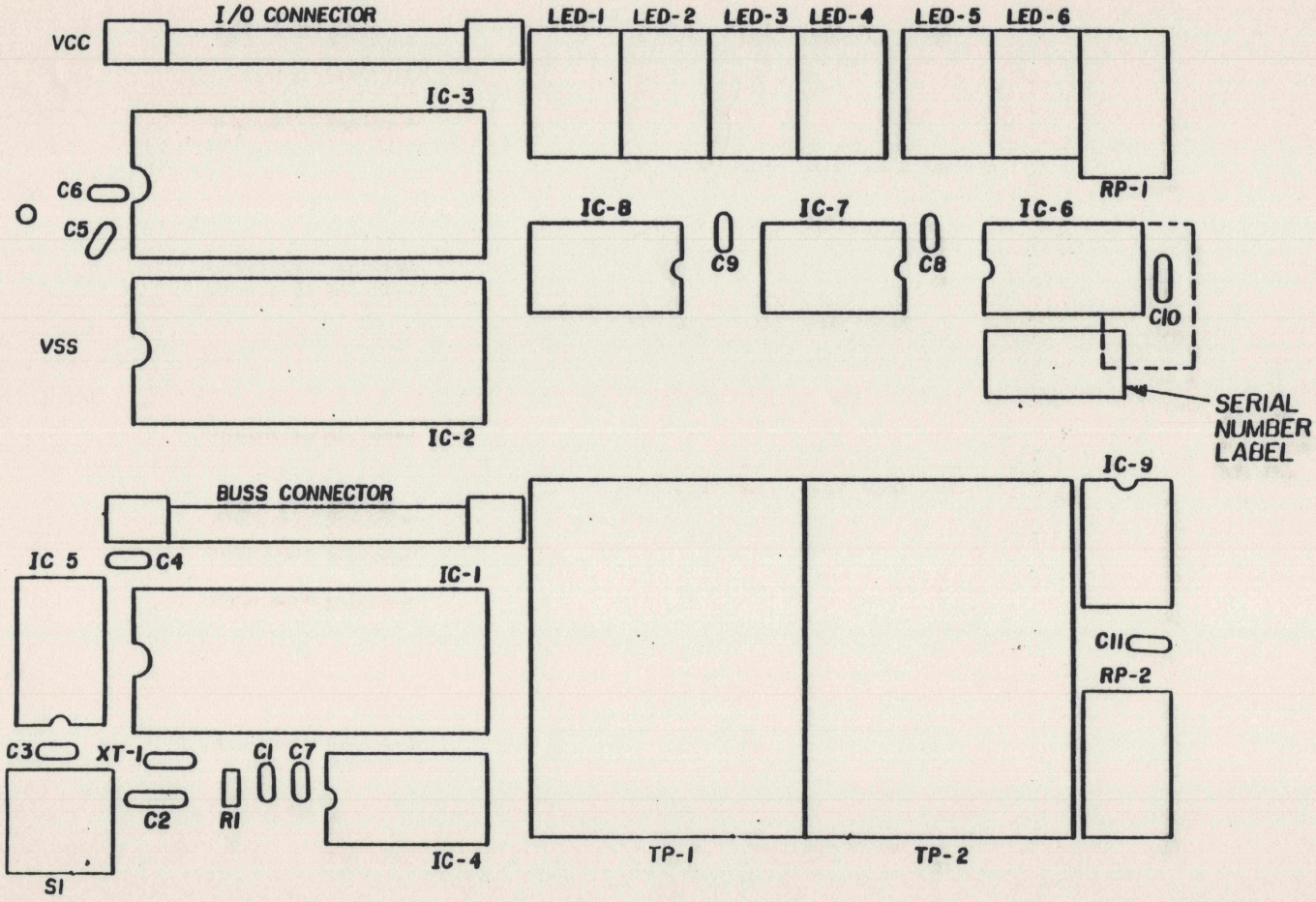

I/O CONNECTOR
VCC
LED-1
LED-2
LED-3
LED-4
LED-5
LED-6
IC-3
RP-1
C6
C5
IC-8
C9
IC-7
C8
IC-6
C10
VSS
SERIAL NUMBER LABEL
IC-2
IC-9
BUSS CONNECTOR
IC 5
C4
IC-1
C11
RP-2
C3
XT-1
C1
C7
C2
R1
IC-4
TP-1
TP-2
S1

FIGURE 1

ASSEMBLY PROCEDURE

Refer to Figure 1 and the PC board to match component designations.

1. IC Socket Installation

Insert each IC socket and align its notch with the notch indicated in the legend for each socket.

a. 40 pin at IC-1, IC-2, and IC-3.

b. 18 pin at IC-4 and IC-6.

c. 16 pin at IC-5, IC-7, and IC-8.

d. 14 pin at IC-9 and all LED's (1 through 6).

Inspect all sockets to make sure each is oriented properly. Solder the sockets to the PC board.

2. Resistor Network

Each resistor network is inserted with its pin 1 occupying the upper left hole of the mounting location.

a. 22Ω resistor pack at RP-1.

b. 3.3KΩ resistor pack at RP-2.

Recheck location of pin 1. Solder resistor packs to PC board.

3. Resistors

Mount flat against the PC board.

a. 22KΩ, 1/4W, 1/2% at R1.

4. Capacitors

a. 20 pF disc at C2.

b. .1 μF at all other locations (C1, C3 to C11).

Solder resistor and capacitors. Trim excess leads.

5. Header Connector

a. Install 34-pin connector at location labeled I/O connector. Open side of the connecter faces the top edge of the PC board.

b. Install optional 34-pin connector at location labeled Buss connector. Open side facing the top edge of the PC board.

Solder header connector(s).

6. Crystal

a. Solder 6.144 MHz at XT-1 (bottom lead only).

b. Solder the short jumper lead from the top of the crystal to the PC board, as shown in Figure 2.

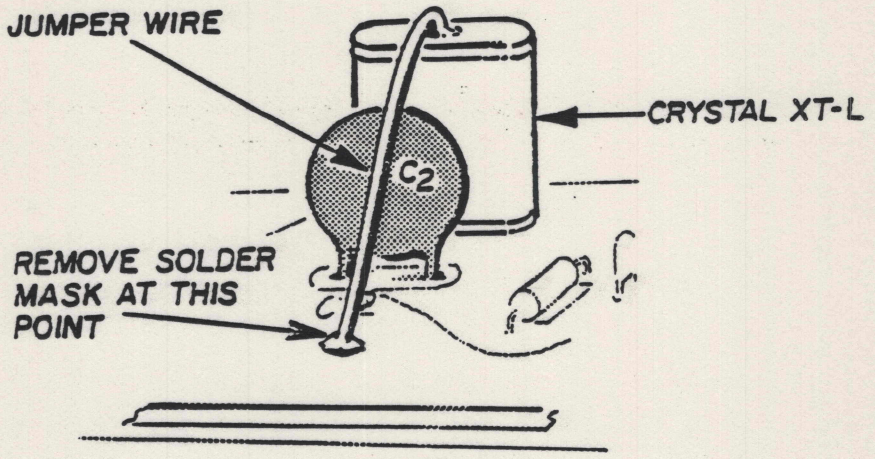


FIGURE 2

7. Push buttons

Insert the components into locations specified and solder.

a. Alpha-numeric keypad to TP-1.

b. Alpha keypad to TP-2.

c. Reset key to S1.

8. Power Terminals

a. Red banana jack at Vcc.

b. Black banana jack at Vss.

Tighten the retaining nut.

9. Protection Circuit

Refer to Figure 3.

a. Connect anode side of diode D1 to the black (Vss) banana jack eyelet.

b. Connect cathode side of diode D1 to the red (Vcc) banana jack eyelet.

c. Connect the negative side of the electrolytic capacitor C12 through the black (Vss) eyelet to the PC board.

d. Connect the positive side of C12 through the red (Vcc) eyelet to the PC board.

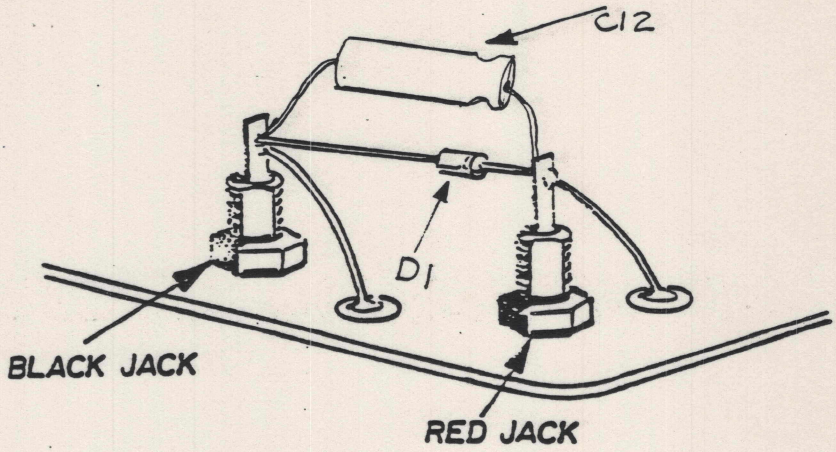


FIGURE 3

10. Installation of IC's

Install IC's into their appropriate sockets with respect to the alignment notches. Be careful not to handle the MOS devices. A static discharge could damage the device.

Component	Location
8085	IC-1
8755	IC-2
8155	IC-3
8185	IC-4
74LS138	IC-5
UDN6118A	IC-6
74LS145	IC-7
74LS138	IC-8
74LS10	IC-9
MAN74A's	LED 1 thru LED 6

11. Final Assembly

a. Apply label to cover.

b. Apply label to inside cover.

c. Align the two mounting clips with the mounting screw holes on the PC board and install on the inside lip of the case.

d. Apply 3M fasteners to bottom of case (if desired).

e. Attach PC board to case with 2 #6 screws.

TEST PROCEDURE

1. Apply +5 Vdc power. If you use a variable supply, be sure to measure the voltage first.

2. Press the RESET button; the display should indicate "- ".

3. Press the keys indicated below and observe the display. Check that all segments light up.

KEY	DISPLAY
W	- --
8	8 --
8	88 --
8	888 --
8	8888 88

4. Run the memory test contained in the Monitor program by pressing R, X, 0500, X or the keys indicated below.

KEY	DISPLAY
RESET	-
X	- --
0	0 --
5	05 --
0	050 --
0	0500 --
X	

The display will blank out for approximately 5 seconds then display "- --" if the memory is good. If the memory is bad it will display "051b P.C.".

OPERATION

FUNCTION KEYS

RESET- separate key in the lower left hand corner. Restarts the monitor.

R- restart, reinitializes the keyboard sequence.

W- write (read) command.

X- store and increment, run, and step.

S- single step.

B- breakpoint.

D- display stack.

P & L- user defined interrupts.

LOADING A PROGRAM OR EXAMINING MEMORY

1. Press R or RESET to reinitialize.
2. Press W to indicate a write command.
3. Key in a 4 digit starting address. The user RAM area is from 1400 to 17FF. The data presently in that location will be displayed.
4. Key in a two hex digit data byte to be stored in that location.
5. Press X to actually enter the data.
6. Repeat steps 3, 4, and 5 until the program has been entered.

RUNNING A PROGRAM

1. Press R or RESET to reinitialize.
2. Press X to indicate a run command.
3. Enter a 4 digit starting address.
4. Press X again to begin program execution.

SINGLE STEP

Single step is used to execute individual instructions for the purpose of debugging a program. One instruction at a time may be executed and then each register may be examined.

1. Press R or RESET.
2. Press S and enter the starting address.
3. Press X. One instruction will be executed. Each successive time X is pushed at this point will display the contents of the registers until all registers have been displayed.
4. Press S to execute a new instruction without examining the registers.

BREAKPOINT

Breakpoints are used similarly to single step except that complete blocks of instructions may be executed. The program will run up to the instruction in which a breakpoint was set. Up to 3 breakpoints may be set. After reaching a breakpoint, the breakpoint is removed.

1. Press R or RESET.
2. Press B. Two dashes and a digit 0, 1, or 2 will appear depending on how many breakpoints have already been set.
3. Enter the address for the breakpoint to occur and press X.
4. Repeat steps 2 and 3 for another breakpoint.
5. Now execute the program as usual. When the breakpoint is reached the program will halt and display the contents of the P.C.